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Department of Computer Science and Engineering | 2026–2027, Odd Semester

Database Systems Engineering and Distributed Backend Development

(25CS1302E) Practical Experiment 1: The Foundation

BookFlow SQL Schema

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Aim

To set up the SQL foundation for a small BookFlow library database by creating the database and Books and Members tables, applying key integrity constraints, inserting sample records, and checking that invalid records are rejected correctly.

Program

File: p1.sql Run using: `psql -U postgres -f p1.sql`

```
-- Practical 1: The Foundation - BookFlow database
-- Student: Chakrika Panakala | Roll No.: 2520090042

DROP DATABASE IF EXISTS bookflow_db;
CREATE DATABASE bookflow_db;
\c bookflow_db

-- Books table
CREATE TABLE Books (
  book_id SERIAL PRIMARY KEY,
  title VARCHAR(200) NOT NULL,
  isbn VARCHAR(13) NOT NULL UNIQUE,
  published_year INT CHECK (published_year < 2027)
);

-- Members table
CREATE TABLE Members (
  member_id SERIAL PRIMARY KEY,
  full_name VARCHAR(100) NOT NULL,
  email VARCHAR(150) NOT NULL UNIQUE
  CHECK (email ~* '^[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,}$')
);

\d Books
\d Members

-- Sample records
INSERT INTO Books (title, isbn, published_year) VALUES
('Atomic Habits', '9780735211292', 2018),
('The Alchemist', '9780062315007', 1988),
('The Pragmatic Programmer', '9780135957059', 2019);

INSERT INTO Members (full_name, email) VALUES
('Chakrika Panakala', 'chakrika.panakala@gmail.com'),
('Meera Shah', 'meera.shah@outlook.com'),
```

```
('Arjun Nair', 'arjun.nair@gmail.com');

SELECT * FROM Books;
SELECT * FROM Members;

-- Constraint checks: these statements should
fail -- 1. Repeated ISBN
INSERT INTO Books (title, isbn, published_year) VALUES
('Another Atomic Habits', '9780735211292', 2020);

-- 2. Missing title
INSERT INTO Books (title, isbn,
published_year) VALUES (NULL, '9781234567890',
2021);

-- 3. Future publication year
INSERT INTO Books (title, isbn, published_year)
VALUES ('Upcoming Book', '9781234567891', 2035);

-- 4. Missing member email
INSERT INTO Members (full_name, email)
VALUES ('No Mail User', NULL);

-- 5. Incorrect email format
INSERT INTO Members (full_name, email)
VALUES ('Invalid User', 'invalid-email');

-- 6. Repeated member email
INSERT INTO Members (full_name, email)
VALUES ('Duplicate User', 'chakrika.panakala@gmail.com');

SELECT COUNT(*) AS total_books FROM Books;
SELECT COUNT(*) AS total_members FROM Members;
```

Output

The following output represents the important results obtained after running the SQL script. The table descriptions confirm that the required constraints were created.

Books table structure

Column	Type	Constraint / Property
book_id	integer	PRIMARY KEY, auto-generated
title	varchar(200)	NOT NULL
isbn	varchar(13)	NOT NULL, UNIQUE
published_year	integer	CHECK (published_year < 2027)

Members table structure

Column	Type	Constraint / Property
member_id	integer	PRIMARY KEY, auto-generated

full_name	varchar(100)	NOT NULL
email	varchar(150)	NOT NULL, UNIQUE
		CHECK for valid email format

Inserted records

book_id	title	isbn	published_year
1	Atomic Habits	9780735211292	2018
2	The Alchemist	9780062315007	1988
3	The Pragmatic Programmer	9780135957059	2019

member_id	full_name	email
1	Chakrika Panakala	chakrika.panakala@gmail.com
2	Meera Shah	meera.shah@outlook.com
3	Arjun Nair	arjun.nair@gmail.com

Constraint Verification

Each test below is intentionally invalid. PostgreSQL should reject the statement because the corresponding constraint prevents inconsistent data from being stored.

Test	Expected reason for rejection
1. Duplicate ISBN	UNIQUE constraint on Books.isbn
2. NULL title	NOT NULL constraint on Books.title
3. Year 2035	CHECK constraint requires published_year < 2027
4. NULL email	NOT NULL constraint on Members.email
5. Invalid email	CHECK constraint validates email format
6. Duplicate email	UNIQUE constraint on Members.email

Representative PostgreSQL errors

```
INSERT ... duplicate ISBN
ERROR: duplicate key value violates unique constraint

INSERT ... NULL title
ERROR: null value in column "title" violates not-null constraint

INSERT ... year 2035
ERROR: new row violates check constraint

INSERT ... NULL email
ERROR: null value in column "email" violates not-null constraint

INSERT ... invalid-email
ERROR: new row violates check constraint
```

```
INSERT ... duplicate email
ERROR: duplicate key value violates unique constraint
```

Final Count

Query	Result
SELECT COUNT(*) AS total_books FROM Books;	3
SELECT COUNT(*) AS total_members FROM Members;	3

Result

The BookFlow database was created successfully with Books and Members tables. Three sample books and three members were added. The validation tests demonstrate that duplicate identifiers, missing required values, invalid email formats, and future publication years are rejected by the defined constraints. The final database contains 3 valid books and 3 valid members.

Conclusion

This practical demonstrates how SQL constraints can be used to maintain data accuracy at the database level. PRIMARY KEY identifies each record, UNIQUE prevents duplicate ISBNs and emails, NOT NULL protects required fields, and CHECK enforces application-specific rules.